

Michael Evans

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☎ (757) 236-1257

EDUCATION

Old Dominion University

M.S. in Electrical & Computer Engineering

Thesis: *AI Framework for Disease-Agnostic Brain Glucose Hypometabolism Prediction*

B.S. in Computer Science

Norfolk, VA

Anticipated May 2027

May 2025

Tidewater Community College

A.S. in Computer Science

Norfolk, VA

May 2023

RESEARCH INTERESTS

Deep learning for medical image analysis, neuroimaging, multimodal learning, neurodegenerative disease modeling

RESEARCH EXPERIENCE

Graduate Research Assistant

Sept 2025-Present

Old Dominion University, advised by Dr. Khan Iftakharuddin

Vision Lab

- Designed and trained a multimodal CNN-Transformer model from scratch to predict regional age-adjusted FDG-PET uptake, enabling hypometabolic profile prediction from s/fMRI scans and CSF proteomics.
- Developed a standardized, BIDS-compliant preprocessing pipeline for multimodal neuroimaging (sMRI, fMRI, FDG-PET) using MATLAB, fMRIPrep, and Clinica, ensuring reproducible research outcomes.

Undergraduate Research Assistant

Sept 2024-Sept 2025

Old Dominion University, advised by Dr. Khan Iftakharuddin

Vision Lab

- Trained a random forest classifier for atypical Alzheimer's disease classification using cognitive test scores and MRI features, improving recall from 34% to 77% on ADNI and 52% to 69% on NACC over baseline.
- Constructed a FreeSurfer pipeline on an HPC cluster to extract regional volumetric MRI features at scale.
- Visualized an ROI-map on an anatomically averaged brain template to highlight brain regions with significant features ($\alpha = 0.05$) for Alzheimer's classification with volumetric group mean z-score color coding.

NSF REU Intern - Developing Self-Drive Algorithms for Electric Vehicles

May-July 2024

Lawrence Tech and Michigan State University, advised by Drs. Chan-Jin Chung and Josh Siegel

Computer Science & Artificial Intelligence Robotics Lab

- Designed an adaptive speed algorithm to reduce acceleration and braking in autonomous vehicles through intersections by up to 75.35%, minimizing fuel consumption and noise pollution.
- Implemented and analyzed DBSCAN and K-means machine learning self-drive algorithms for reliability.
- Built a simulation in GazelleSim for testing our autonomous intersection control in a controlled environment.

NSF REU Intern - Disinformation Detection and Analytics

June-Aug 2023

Old Dominion University, advised by Dr. Jian Wu

Lab for Applied Machine Learning and Natural Language Processing Systems

- Engineered prompts and tuned hyperparameters with the OpenAI API for scientific claim verification.
- Increased the size of our project corpus by over 200% to improve benchmarking the effectiveness of the GPT-3.5-turbo model's ability to generalize across multiple scientific news domains.
- Calculated the precision, recall, and F1 score on zero-shot classification with the GPT-3.5-turbo model against our dataset on two sub-tasks: stance labeling and identifying sentence rationales.

PUBLICATIONS

Preprints

P1. M. A. Witherow, M. L. Evans, A. Temtam, H. R. Okhravi, and K. M. Iftakharuddin. Machine Learning-Enhanced Non-Amnesic Alzheimer's Disease Diagnosis From MRI and Clinical Features. *arXiv* 2026.

Peer-Reviewed Papers

C2. M. L. Evans, M. Machado, R. Johnson, L. Escamilla, A. Vadella, B. Froemming-Aldanondo, T. Rastoskueva, M. Jostes, D. Butani, R. Kaddis, C. Chung, and J. Siegel. A Roadside Unit for Infrastructure Assisted Intersection Control of Autonomous Vehicles. *IEEE International Conference on Electro/Information Technology (EIT)* 2025.

C1. B. Froemming-Aldanondo, T. Rastoskueva, M. L. Evans, M. Machado, A. Vadella, L. Escamilla, R. Johnson, M. Jostes, D. Butani, R. Kaddis, C. Chung, and J. Siegel. Evaluating Low-Resource Lane Following Algorithms for Compute-Constrained Automated Vehicles. *IEEE International Conference on Artificial Intelligence, Robotics, and Control (AIRC)* 2025.

W1. M. L. Evans, D. Soós, E. Landers, and J. Wu. MSVEC: A Multidomain Testing Dataset for Scientific Claim Verification. *National Workshop for REU Research in Networking and Systems (REUNS @ ACM MobiHoc)* 2023.

SELECTED PROJECTS

Resting State Functional MRI Based Alzheimer's Disease Classification

ECE 784: Computer Vision Term Project

- Trained an SVM with SelectKBest feature selection to classify Alzheimer's disease from resting-state fMRI data on an ADNI cohort of 66 patients, achieving up to 0.71 balanced accuracy.
- Performed rs-fMRI preprocessing and AAL2 brain parcellation to extract: functional connectivity, amplitude of low-frequency fluctuations (ALFF), regional homogeneity (ReHo), and mean BOLD signal features.

Pre-Processing and Noise Distribution Learning in MRI

ECE 783: Digital Image Processing Term Project

- Applied non-local means filtering to denoise MRI corrupted by additive Rician noise, achieving a 39.8% SNR increase and experimentally verifying that the Rician distribution approaches Gaussian at high SNR.
- Modeled MRI acquisition noise by generating Rician samples from a bivariate normal distribution, and performed residual distribution analysis between the clean reference image and denoised images.

SELECTED TECHNOLOGIES

Programming Languages

Python, R, Bash, C++, MATLAB

Libraries & Frameworks

Medical Imaging: fMRIPrep, Clinica, FreeSurfer, SPM, NiBabel, Nilearn

Machine Learning / General: PyTorch, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

AWARDS

NSF CISE REU Travel Grant – \$1,200	May 2025
Research Profile Highlight – Old Dominion University College of Sciences Newsletter	Nov 2024
College of Sciences Dean's List – Old Dominion University	Dec 2023-May 2025
BIO, GEO, CISE and OCE REU Travel Grant – \$1,200	Sept 2023
Academic Excellence Award – Tidewater Community College	April 2020, Mar 2023

SERVICE & MEMBERSHIPS

Congressional App Challenge (CAC), <i>Vision Lab Exhibitor</i>	Dec 2024
Intelligent Ground Vehicle Competition (IGVC), <i>Surveyor</i>	Jun 2024
Institute of Electrical and Electronics Engineers (IEEE), <i>Student Member</i>	Jun 2024-Present
Association for Computing Machinery (ACM) Group at ODU, <i>Member</i>	Jan 2024-Present

ADDITIONAL EXPERIENCE

PHP Web Developer	Dec 2021-June 2022
Hard to Find Party Supplies	